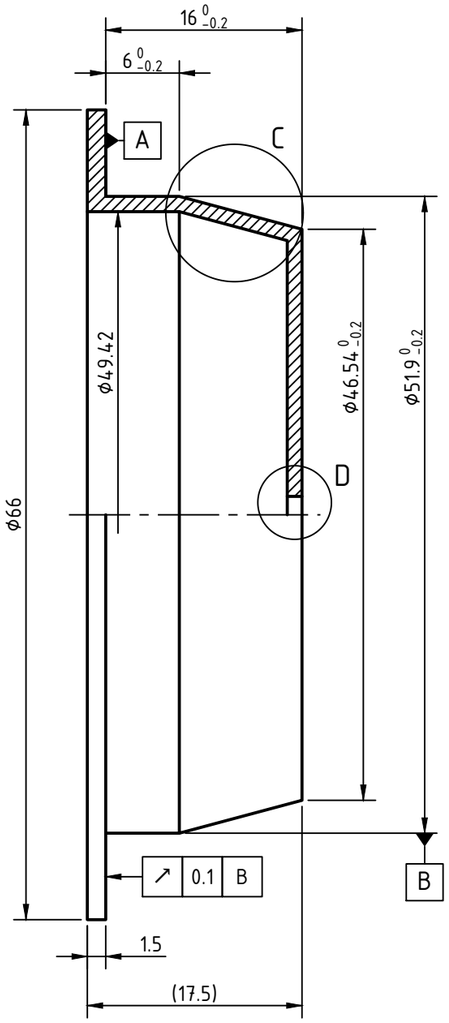
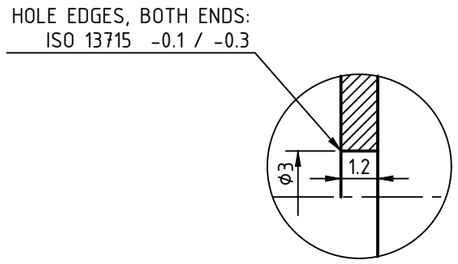
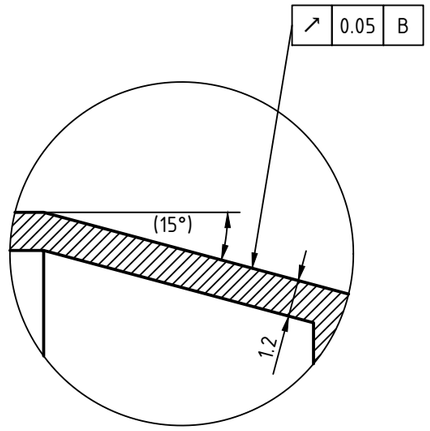


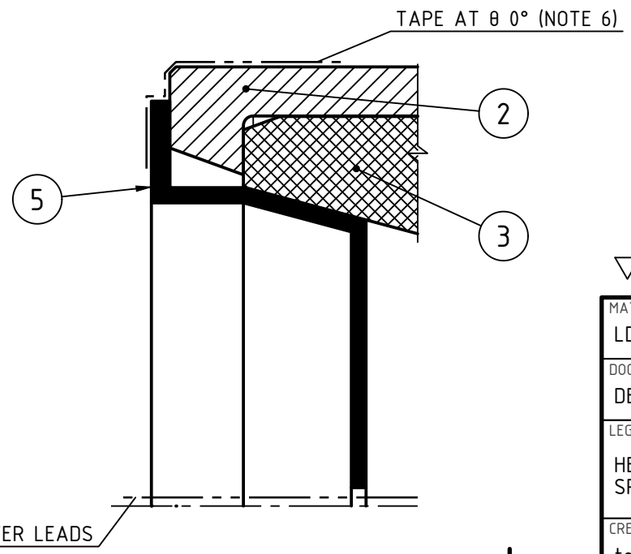


DETAIL D (5:1)



DETAIL C (5:1)

INSTALLATION (REFERENCE) 2:1



NOTES

1. Material: low-density polyethylene PE-LD (ISO 17855), natural, unfilled, unpigmented, no regrind; density about 0.92 g/cm³. Extruded rod ϕ70. Certificate of conformity.
2. Turned from rod, one cap per firing. Op 1: face the aft end, bore ϕ49.42 and the inner cone to the 1.2 end wall, drill ϕ3, turn ϕ66, part off long. Op 2: on a turned mandrel in the bore, turn the cone, ϕ51.9, seat face A and the plug end to length. Sharp high-rake tools, light finishing cuts, no oil. The geometry also suits injection moulding (no undercuts, core drawn aft).
3. Fit: slip fit in the 15° graphite divergent of insert SRM100-104, 0.05 radial clearance at maximum material. All size and station tolerances on the plug go towards clearance: at worst it touches the graphite (profile 0.1) along a line and never wedges; a 15° cone is not self-locking and LDPE cannot score graphite. Seat face A bears on the aft face of holder SRM100-102, but the graphite (and its divergent) is placed by the holder's lip, 6 +0.1/0 (SRM100-102): the clearance holds only with that tolerance. Each 0.1 the lip is thinner moves the divergent 0.027 radially onto the plug; 0.19 thin gives contact. ϕ51.9 clears the lip bore ϕ54 by over 1 mm and the plug clears the graphite's 45° exit break (SRM100-104, stations 6-7).
4. Measure at 20 ± 2 °C (LDPE grows about 0.01 per K on ϕ52). Runouts on the Op 2 mandrel. Inspect 100 %: plug diameters, stations 6 and 16 from A, walls, ϕ3; trial fit in the holder with the insert fitted: enters by hand, flange seats with no rock, lifts out by hand. Visual: no burrs, swarf, cracks or dirt.
5. Edges: both ends of the ϕ3 lead hole ISO 13715 -0.1 / -0.3, no burr (it would cut the lead insulation); other edges per the title block. No marking on the part: bag each cap and label the bag SRM100-106 A.
6. Fitting (SRM100-000 sheet 1 note 4; README, static-fire procedure): fitted at assembly as the debris seal (taped per this note) and kept on through storage and transport. At the stand it comes off only to insert the igniter (leads shunted, pushed through the throat about 615 mm to the middle of the forward grain's core, out through the cap's ϕ3 hole from the plug end), then it is refitted, flange on the holder, and re-taped with new tape. The igniter always goes in last, at the stand: never move a motor with an igniter in it. Tape: two strips of PVC electrical tape, 19 wide, centred at θ 0° and 180°, 90° from the temperature labels at θ 90° and 270° (SRM100-102 note 8), from the flange's aft face over its rim and the holder's aft face onto its aft body ϕ71.58, the ends at most 15 long there; never over a label, the cup opening or the leads.
7. Removal: the cap blows out at ignition (a moisture and debris seal, not a pressure closure). Misfire: disarm and wait at least 60 s (5 min advised for an N motor) before approaching; cut the tape, pull the cap off along the leads, then remove the igniter. Single use: scrap the cap after every firing or misfire; never re-use it. After firing, once the temperature labels have been read and recorded (before extraction, SRM100-102 note 15), remove the tape residue from the holder with IPA.
8. Store bagged, clean and dry, away from solvents, heat and sunlight.
9. Used on SRM100-000 item 5 (nozzle cap: fitted at assembly as the debris seal, refitted at the stand over the igniter leads). Item numbers in the installation view per SRM100-000.

Ra 3.2

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
LDPE natural, unfilled, rod ϕ70	6.6 g	ISO 2768-mK	Ra 3.2	ISO 13715 -0.3 / +0.3

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED - FOR REVIEW	2:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	NOZZLE CAP Moisture / debris cap - LDPE plug in the 15° divergent

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	-	2026-09-23	SRM100-106	A	1 / 1